Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Knowledge organiser

Learning goal

I can use continuous-sampling evidence to investigate how pH affects amylase activity.

Read a tested sample

Continuous sampling means testing separate small samples at fixed intervals while the reaction continues. Blue-black means starch detected; orange-brown remaining means starch not detected in that drop.

Keep the comparison valid

Change pH; measure endpoint time / s. Match temperature, enzyme and starch amounts, volumes and the endpoint rule. Prepare labelled iodine wells before mixing; keep iodine out of the main reaction.

State the timing limit

The 30 s interval gives a window, not an exact midpoint. If the 60 s sample were missing, 30 s blue-black and 90 s orange-brown would give a 60 s window. Preserve a missed sample or uncertain colour.

Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

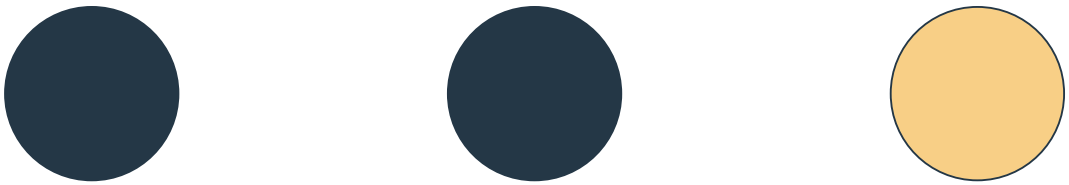
Knowledge organiser · continued

Use a bounded conclusion

First-series endpoint times: pH 4 = 240 s; 5 = 180 s; 6 = 120 s; 7 = 90 s; 8 = 150 s. pH 7 is fastest in this illustrative series. One trial per pH does not prove an exact universal optimum.

Diagram

Dark blue shows blue-black iodine. Orange shows iodine that stays orange-brown. The words give the colour result too.



30 s	60 s	90 s
Blue-black	Blue-black	Orange-brown

Supplied example: samples every 30 s.